Associatively Segmenting Semantics and Estimating Height From Monocular Remote-Sensing Imagery

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***Abstract*— Numerous deep-learning methods have been successfully applied to semantic segmentation (SS) and height estimation (EH) of remote-sensing imagery. It has also been proved that such a framework can be reusable for multiple tasks to reduce computational resource overhead. However, there are still some technical limitations due to the semantic inconsistency between 3-D and 2-D features and the strong interference of different objects with similar spectral–spatial properties. Previous works have sought to address these issues through hard (HPS) or soft parameter sharing (SPS) schemes. But due to unintentional integration, the specific information transmitted between multiple tasks is not clear or in a lot of redundancy. Furthermore, tuning the weights by hand between classification and regression loss functions is challenging. In this article, a novel multitask learning (MTL) method, termed associatively segmenting semantics and estimating height (ASSEH), is proposed to associatively segment semantics and estimate height from monocular remote-sensing imagery. First, considering semantic inconsistency across tasks, we design a task-specific distillation (TSD) module containing a set of task-specific gating units (TSGUs) for each task at the cost of fewer parameters. The module allows for task-specific features to be tailored from the backbone while allowing for task-shared features to be transmitted. Second, we leverage the proposed cross-task propagation (CTP) module to construct and diffuse the local pattern graphlets at the common positions across tasks. Such a high-order recursive method can bridge two tasks explicitly to effectively settle semantic ambiguities caused by similar spectral characteristics with less computational burden and memory requirements. Third, a dynamic weighted geometric mean (DWGeoMean) strategy is introduced to dynamically learn the weights of each task and be more robust to the magnitude of the loss function. Finally, the results of the ISPRS Vaihingen dataset and the urban semantic 3-D (US3D) dataset well demonstrate that our ASSEH achieves state-of-the-art performance.**

***Index Terms*— Height estimation (EH), multitask learning (MTL), remote-sensing imagery, semantic segmentation (SS).**

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I. Introduction

BOTH segmenting semantics and estimating height aim to interpret 2-D contextual and 3-D geometric information regions represented by sets of minimum units in the scenes, where each pixel corresponds to a class of stuff and a height value. The two tasks are related and fundamental, yet challenging in the remote-sensing scenes, such as urban planning, building extraction, military reconnaissance, and so on. With the rapid development of airborne optical images and spaceborne optical images, generous research has been done on semantic segmentation (SS) and height estimation (EH). Although excellent performance has been made for every single task, there are few methods to tackle these two tasks in association.

Traditionally, SS and EH are tackled in isolation, and the interaction between tasks is rarely considered. For SS, the recent efforts [1]–[3] employ deep convolution neural networks (DCNNs) to obtain a powerful feature representation. However, parsing the semantic regions with similar 2-D appearance with remote-sensing optical images exclusively is still very challenging. A certain amount of works [4]–[6] are devoted to considering the statistical characteristics of the heterogeneous data and investigating the multimodal feature fusion strategy. However, these methods require accurate digital surface model (DSM) data as additional input. How to alleviate the data constraint and use multimodal information at the same time to assist SS is still a challenge. For EH, the recent efforts [7]–[9] adopt DCNNs to extract geometric information from monocular remote-sensing imagery. Unfortunately, a remote-sensing image may correspond to multiple real scenes, that is, mapping intensity or color from monocular vision to a height value results in an inherent ambiguity. There are many technically ill-posed problems exploiting monocular cues alone, where each height value is physically credible in the real-world applications.

It has been well recognized both challenges can be addressed cohesively by introducing a unified approach to integrate the 2-D semantic analysis ability of the SS method and the 3-D elevation prediction ability of the EH method. We rely on the assumption that changes in semantic categories usually correspond to transitions in height values and vice versa. Compared with single-task learning (STL), multi-task learning (MTL) cannot only improve inference speed and reduce memory consumption, but also share information visual features among related tasks. At the same time, one can be acted as a regularizer for one another to improve task

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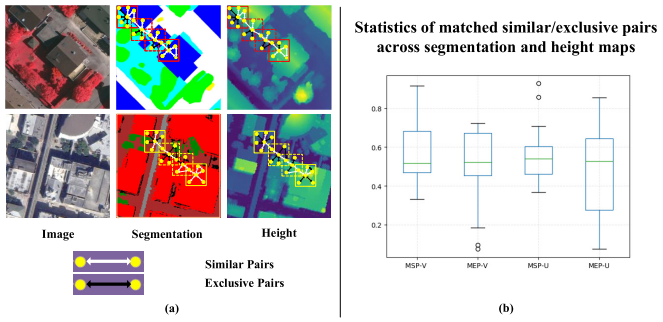


Fig. 1. Statistics of matched similar/exclusive pairs across segmentation and height maps. (a) Common local pattern visual exhibition at multitask labels level. The graphlet in the box is established based on local areas across two tasks, where the points colored yellow represent pixels. The points connected with the white line indicate similar pairs, whereas the points connected with the black line indicate exclusive pairs. (b) Boxplot for the success rate of matched similar/exclusive pairs, where MSP-V and MSP-U separately indicate matched similar pairs in Vaihingen and urban semantic 3-D (US3D) datasets, while MEP-V and MEP-U indicate matched exclusive pairs. By calculating the success ratio of matched pairs in Vaihingen and US3D datasets, it is found that the success rate of matched pairs is quite high, but there is also a scenario where similar or exclusive pairs fail to match.

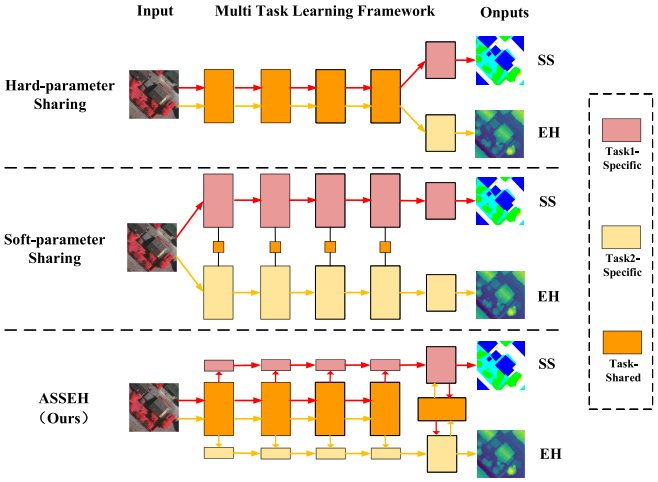


Fig. 2. Conceptual overview for our approach. Given a remote-sensing MTL spectacle involving SS and EH. Traditionally, HPS stacks task-specific layers on top of the task-shared layers. SPS learns a task-specific subnetwork for each task, but each subnetwork can access the information inside the other. In contrast, our associatively segmenting semantics and estimating height (ASSEH) can automatically learn the task-shared and task-specific features at different stages.

performance. In addition, the STL of SS only interprets the 2-D scene, while joint learning of SS and EH can help us better understand the 3-D scene interpretation. Recently, MTL approaches [10]–[13] have sprung up and have been widely applied in numerous visual tasks, including semantic labeling, normal, and depth estimation. The existing MTL networks can be divided into hard parameter sharing (HPS) [12], [14], [15] and soft parameter sharing (SPS) [11], [16], [17] schemes. The former is that all tasks share the same backbone network and perform learning directly from the shared feature pool for each task. This method effectively reduces the number of network parameters, but the generalized representation is easy to lead to under-fitting. The latter is that each task has its own parameters and encourages parameter similarity by constraining the differences between the parameters of different tasks. Although this method greatly improves the multitask performance, the model parameters will increase linearly with the increase of the number of tasks, which is not conducive to practical application. In addition, numerous studies [18]–[20] have been devoted to balancing the multitask joint learning to avoid the situation where a certain task has a dominant impact on network weights.

However, MTL methods of natural images cannot be directly applied to remote-sensing images, due to the strong interference of cross-task semantic inconsistency and the complex spectral–spatial properties of remote-sensing orthophoto images. The former mainly refers to those objects belonging to the same category that may have different height values and vice versa, and the latter indicates that there are different objects with similar spectral properties (e.g., road and roof, tree and lawn) and shadow occlusion in orthophoto images. Inspired by [12], we explore explicitly the second-order patterns at the common positions between SS and EH in remote-sensing scenes, which contains similar and exclusive relation among pixels. As shown in Fig. 1(a), there exist some common similar pairs (connected with white line) and exclusive pairs (connected with black line) in different tasks. Here, we employ label consistency to measure the similarity among pixels in segmentation maps and apply the average relative error (Rel) whose threshold is set to 0.3 as the metric in height maps according to the state-of-the-art performance in [8] and [9]. Due to the limited resources and memory, we crop the 512×512 labels into 16 nonoverlapping patches and draw the boxplot based on the statistical results of small patches. During the statistical process, we find that similar or exclusive pairs fail to match in some scenarios, which further verifies that there are semantic inconsistencies between the two tasks in these scenarios. However, as illustrated in Fig. 1(b), the statistical results about the success ratios of matching pairs across tasks in the Vaihingen and urban semantic 3-D (US3D) datasets are quite high, where more than 50% similar pairs and exclusive pairs are matched successfully. This indicates that there are a large number of pattern-affinitive pairs across two tasks, and the patterns can be diffused between tasks to enhance the discrimination for different objects with similar spectral–spatial properties. Therefore, we may conclude that associatively performing SS and EH from monocular remote-sensing imagery should not only pay attention to task-specific features for each task, but also focus on cross-task propagation (CTP) explicitly. In addition, automatically and efficiently learning the weights for each task has not yet been well developed in the study of multitask pixel prediction from remote-sensing images.

In order to address the above-mentioned issues, a novel MTL approach is proposed to associate SS and EH, termed associatively segmenting semantics and estimating height (ASSEH) (see Fig. 2). First, considering semantic inconsistency across tasks, we propose a task-specific distillation (TSP)

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module, which applies a task-specific gating unit (TSGU) at each convolution stage in the shared network to autonomously distinguish the importance of task-shared features to their respective tasks in a self-supervised manner. This efficient and flexible strategy enables task-shared features to be generalized across tasks while ensuring task-specific features can be customized for each individual task. Compared with the MTL approaches [11], [16] with explicit separation representation, our architecture has fewer parameters to capture the task-shared and task-specific features at the same time. Second, for the complex spectral–spatial properties of remote-sensing orthophoto images, we introduce a CTP module through explicit cross-task interactions to effectively settle semantic ambiguities caused by similar spectral characteristics. Inspired by [21] that adjacent pixels share most of their neighbors, we further explore higher-order common patterns across two tasks. A cross-window affinity learning (CwLAL) layer is applied to construct local pattern graphlets, which record the correlation of pixel-level patterns in a local region. We derive the higher-order affinitive patterns at the common positions and then mutually transfer the counterpart graphlets across the two tasks by introducing a modality-affinitive diffusion (MAD) module. To capture the long-range context, we adopt multiple recursive diffusions to reduce memory requirements as well as computation burden. Different from [13], the CTP strategy does not construct graphlets around each pixel but uses adjacent pixels to share a local graphlet and build the cross-window connections to enlarge receptive fields with less reduced computation burden. Third, for the multitask objective function, we propose a novel and simple dynamic weighted geometric mean (DWGeoMean) strategy to automatically learn the weights for each task and be more robust to the magnitude of the loss function. Additionally, a postprocessing method is presented to combine the small patches to generate a seamless continuous height map.

In summary, the main contributions of this article are summarized as follows.

1) Different from the most STL approaches, we present a novel MTL framework, termed associatively segmenting semantics and estimating height (ASSEH), to associatively segment semantics and estimate height from monocular remote-sensing imagery.
2) Considering semantic inconsistency across tasks, we design a TSD module to allow the task-specific features to be tailored from the backbone and the task-shared features to be transmitted across tasks. It benefits from the task-shared features for generalization across tasks and the task-specific features to avoid under-fitting.
3) To alleviate semantic ambiguities caused by similar spectral characteristics, an efficient CTP module is proposed to mine common local patterns across tasks by constructing and diffusing the high-order graphlets.

The rest of this article is organized as follows. Section II details the STL and MTL for SS and EH. The details of our proposed method are illustrated in Section III. Section IV provides the experimental evaluations as well as the corresponding results analysis. Finally, Section V draws some conclusions.

II. Related Works

A. Single-Task Learning

With its strong feature representation ability, convolutional neural networks (CNNs) have made great progress in many computer vision tasks. The typical approach is to train a separate network for each specific task and only focuses on the optimization of the task metrics, which is uniformly called the STL method. In this article, we focus on two tasks: SS and EH.

1) Semantic Segmentation: Aims to parse the scene via assigning categories pixel by pixel. Conventional segmentation algorithms [22] only utilize the similarity between pixels in low-level features to divide the image region, resulting in less semantic information in the segmentation results. Over the past decade, more and more CNNs have been proved to bring impressive results to SS. Long *et al.* [23] realize the full convolution network (FCN) for the first time by changing the last three full connection layers of the network into convolution layers, which solves the problem of image segmentation at the semantic level. In addition, FCN combines low-level detail information and high-level semantic information by introducing a skip architecture to ensure the robustness and accuracy of the network. This work can be regarded as a milestone for the method of SS. Since then, more and more frameworks based on FCN have been designed to explore contextual modeling. PSPNet [24] extracts long-range dependencies at different scales by proposing a pyramid pooling module (PPM). Chen *et al.* [25] propose an atrous spatial pyramid pooling (ASPP) module to mine multiscale features to capture the serviceable context information. Moreover, inspired by nonlocal network [26], several frameworks have been designed to gather global spatial information in a manner of self-similarity between pixels, such as DANet [27], OCNet [28], and CCNet [29].

For remote-sensing image classification, numerous advanced methods based on encoder–decoder architecture have been proposed to obtain more accurate segmentation results. Mou *et al.* [1] model the long-distance spatial relationship in remote-sensing images to enhance the expression ability of image features by introducing spatial as well as channel relation modules. Niu *et al.* [3] introduce a remote-sensing image SS method based on hybrid attention mechanism. Considering the inherent category differences between remote-sensing image spectra, the work explicitly modeled the similarity of feature categories and channels. At the same time, it proposed a region shuffle attention module to reduce the algorithm complexity of the attention mechanism. To settle ambiguities by introducing additional elevation data, Sherrah [4] considered the different statistical characteristics of the two modal data and fine-tune pretrained DCNNs on the remote-sensing data to make better use of image features. Audebert *et al.* [5] introduced a multikernel convolutional layer for semantic aggregation and adopted a residual correction for heterogeneous data fusion. In the follow-up

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work [6], they systematically investigated the early and late fusion of multimodal data (e.g., Lidar and multispectral data) to deal with semantic labeling based on multimodal remote-sensing data. In order to make the best of elevation data to assist SS and alleviate the existence of over-smooth during global context aggregation methods, Liu *et al.* [30] propose a height-aware method to jointly reason the contextual as well as geometric information, while adaptively allocate the global information to each pixel position by introducing a context-aware reorganization module.

In this work, we consider introducing geometric information to improve the performance of SS. Moreover, most existing methods develop a two-stream network with an accurate height map as additional input to incorporate multimodal information, which severely limits their applications. In this article, we jointly infer the semantic and height information from monocular remote-sensing images for height-aware SS.

2) Height Estimation: Aims to parse the scene by capturing the spatial geometric information pixel by pixel, which is regarded as one of the pivotal components of DSM. Previous researches on EH from remote-sensing images mostly focus on multiview photogrammetry [31], stereo-pair photogrammetry [32], and structure from motion. These methods have very strict requirements for the input data, which need to provide two or more images from different angles or overlapping stereo pairs and need a series of preprocessing as well as postprocessing for the data. Thus, a straightforward idea is estimating the height from monocular inputs. With the popularity of CNNs, some works attempt to estimate depth by introducing deep convolution networks and have achieved considerable performance. Eigen *et al.* [33] predict depths by proposing a multiscale neural network, which consists of coarse-scale and fine-scale networks to predict depth more accurately. The authors in [34] incorporate CNNs with probabilistic graphic models, in which the conditional random fields (CRFs) are utilized to refine the relationship between adjacent superpixels. In addition, some literature [35], [36] propose ordinal regression algorithms based on classification algorithms to learn a rule to predict depth. In the field of remote sensing, a certain amount of work attempts to adopt CNNs to estimate height from monocular aerial or satellite images. Mou *et al.* [7] propose an encoder–decoder network with a skip connection and a regression loss to predict height map from optical images. Ghamisi and Yokoya [37] propose an encoder–decoder framework based on an adversarial network to simulate DSM. Similar encoder–decoder networks are also adopted for height regression prediction in [8] and [9].

The existing state-of-the-art models obtain the corresponding height maps by only extracting geometric information, which is extremely challenging as a result of the lack of abundant contextual information. In this article, we jointly infer the semantic and height information from monocular remote-sensing images for semantic-aware EH.

B. Multitask Learning

MTL aims to improve the generalization performance of the model by sharing and transferring the serviceable information contained in multiple tasks. As illustrated in Section I, these MTL networks bring numerous advantages compared with STL. Particularly, geometric and semantic information is naturally stored in optical images and can be captured from monocular remote-sensing imagery. Therefore, more and more MTL networks have been applied to dense prediction tasks such as depth estimation [13] and SS [12].

1) Multitask Architecture: Aims to learn a generalizable representation among different tasks and also provide specific features for each task, which mainly includes encoder- and decoder-focused approaches. The former mainly relies on the encoder to learn task-shared features among different tasks and then adopt the task-specific heads to process these features separately. Cross-stitch networks [16] propose cross-stitch units to linearly combine task-shared and task-specific activation features. Inspired by [16], Gao *et al.* [39] fuse the activation maps from different tasks in the encoding stage by employing a dimensionality reduction mechanism. In addition, MTAN [11] applies a soft attention module for each task to automatically extract the task-specific features. The above methods directly predict all task outputs via task-specific heads, which fail to capture differences and relationships across tasks. Different from encoder-focused frameworks, decoder-focused frameworks first adopt a multitask encoder for initial predictions and then share or transfer information in the decoding phase to further improve the performance of each task. PAD-Net [40] designs a multitask guided prediction-and-distillation network with multimodal distillation modules to incorporate multimodal information for simultaneous SS and depth estimation. In addition, PAP-Net [12] proposes task-specific and CTP to integrate affinity patterns, which recur much frequently among tasks. Zhou *et al.* [13] improves the pattern affinities to distill and separate both intertask and intratask patterns for MTL.

Our approach is different from the above methods in the following aspects: 1) in the encoding part, in addition to the backbone sharing network, we design a TSP module at each stage, which allows automatically learning task-shared and task-specific features at the cost of fewer parameters; and 2) in the decoding part, we mine cross-window and cross-task local affinity patterns and diffuse them from the local region to the global region. Compared with the previous methods, our method has the advantage of efficient calculation and faster iteration.

2) Multitask Optimization: Aims to balance the multitask joint learning and avoid the situation that a certain task has a dominant impact on the network weight. Yet, the general, the objective function in MTL can be expressed as in an arithmetic mean weighted manner

$$\mathcal{L}_{\mathrm{MTL}} = \sum_i w_i \mathcal{L}_i \tag{1}$$

where $\mathcal{L}_i$ indicates the loss for each task and w_i is the weight for each loss. Assuming using the stochastic gradient descent (SGD) to optimize the objective function in (1), the network weights $\mathcal{W}^+$ during back-propagation can be continuously updated according to the following formula:

$$\mathcal{W}^+ = \mathcal{W} - \eta \sum_i w_i \frac{\partial \mathcal{L}_i}{\partial \mathcal{W}}. \tag{2}$$

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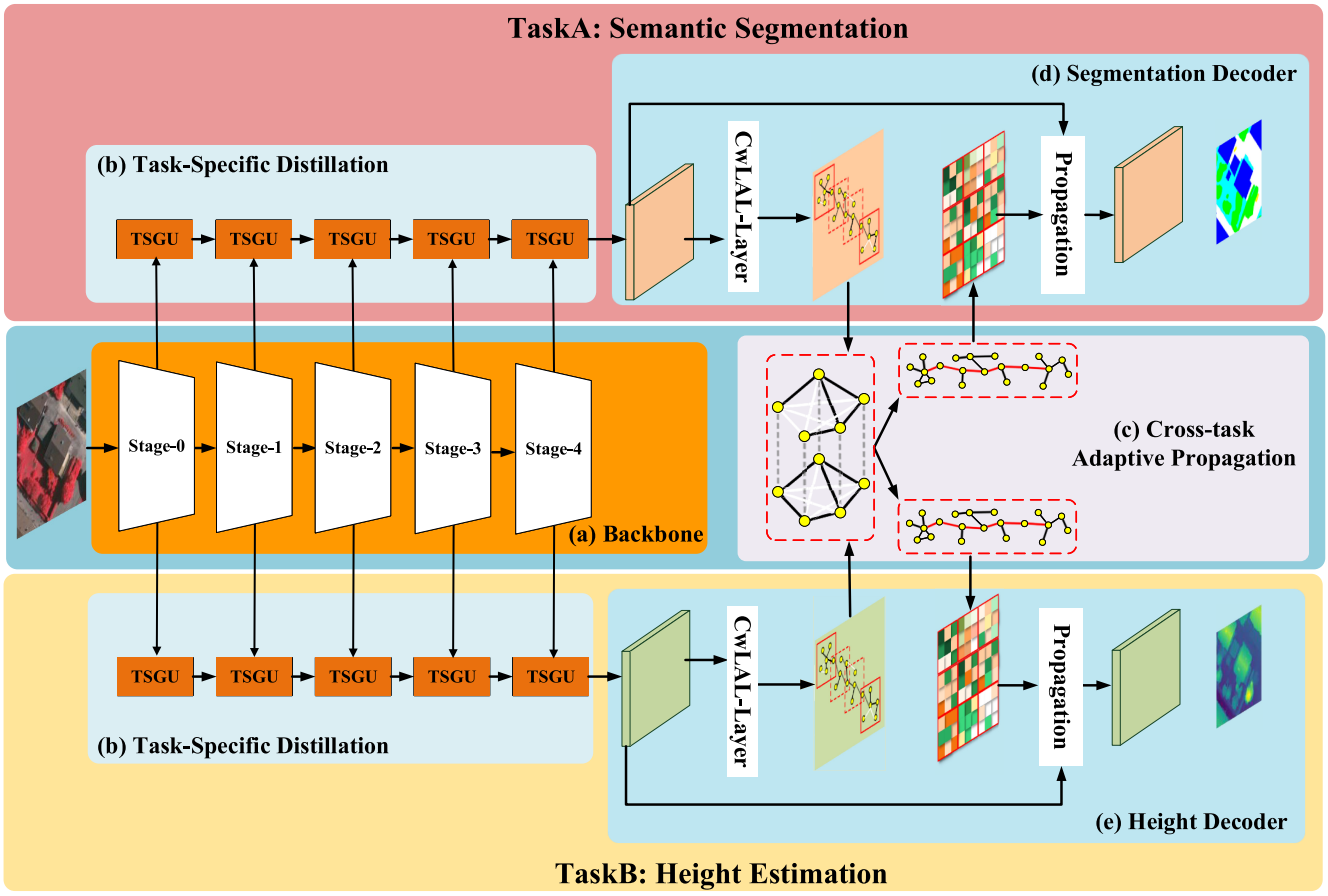


Fig. 3. Structure of the proposed ASSEH method for joint prediction of segmentation and height maps. (a) Backbone, which is based on Resnet [38] to extract features. (b) TSP embraces a set of TSGUs at different stages. (c) Cross-task Adaptive Propagation, which is designed to construct and diffuse the local pattern graphlets across tasks. (d) Segmentation decoder to predict segmentation maps. (e) Height decoder to predict height maps.

From (2), we can draw the conclusion that each MTL method cannot only adjust the task-specific weights w_i, but also adapt the task-specific gradients ($\partial \mathcal{L}_i / \partial \mathcal{W}$) to balance the weight update of MTL network.

For task-specific weights, Kendall *et al.* [18] consider the homoscedastic uncertainty to weight multiple loss functions, which aims to give higher weight to simple tasks. In addition, MTAN [11] records the loss values of different steps to build task-specific weights, which aims to learn at a similar speed for each task. For task-specific gradients, GradNorm [19] presents a gradient normalization algorithm to dynamically tune gradient magnitudes, in which different tasks learn at similar speeds. Sener *et al.* [20] consider the MTL as multiobjective optimization and present an upper bound to yield a Pareto optimal solution.

In summary, these strategies focusing on balancing the speed of multitasking learning, either explicitly define the loss function or require accurately optimizing the gradient descent among tasks. In this work, we propose a principled strategy with minimal design complexities for combining multiple objective functions to consider the learning speed and the magnitude of loss function for each task.

III. Methodology

STL approaches refer to designing an independent network for different tasks. The aim of this work is to improve the performance of a single task by employing MTL frameworks. Accordingly, we propose a cooperative framework for both SS and EH tasks. This part first overviews the whole frame structure in Section III-B and then further introduces the details of TSP and CTP in Section III-C. Finally, Section III-D poses the model objective with the DWGeoMean strategy for two joint dense prediction tasks.

A. Overview

As shown in Fig. 3, the proposed MTL framework is encapsulated into encoder–decoder structure. The whole network consists of five components: a shared backbone to extract image features, TSP modules embracing a set of TSGU for each task, a CTP module to construct and diffuse the local pattern graphlets across tasks, and two task-specific decoders to predict segmentation and height maps.

Concretely, given an image X, we first pass X through a shared backbone ResNet-101 [38] to produce multiscale hierarchical features. We feed each hierarchical feature map into a TSGU to filter the features required by the specific task. Then the feature mapping of the last TSGU in the TSP stream is fed into each task-specific decoder to capture task-related information. Then we execute CTP on the two task-specific features to optimize the final prediction results. The whole process benefits from both cross-window local affinity learning (CwLAL) for task-specific context learning and MAD

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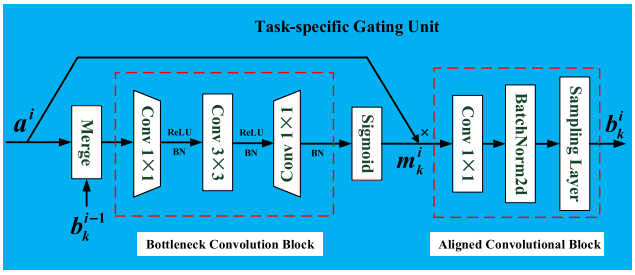


Fig. 4. Details of the TSGU, which consists of a bottleneck convolution block and an aligned convolution block. Here, a^i represents the shared feature in stage i of the task-shared backbone, and ${b_k}^i$ indicates the specific feature of the task k at the stage i.

for cross-task commonalities interaction. Finally, the whole network can be optimized by a new objective function with the DWGeoMean strategy to solve the task balancing problem. The details of the TSP, CTP, and DWGeoMean strategy will be illustrated in the following sections.

In addition, a postprocessing method is presented to combine the small patches to generate a seamless continuous height map. Since the cropped patches are normalized in the training stage, the predicted height map is localized. The results of connecting adjacent small height maps directly to produce a large heightmap can cause gaps between adjacent patches. To alleviate the issue, we first employ a 2-D Gaussian smoothing kernel function ($\mu = 0, \sigma = 0.2$) to adjust the swing of the height value and shorten the height difference among adjacent patches. Then, we splice the adjacent patches into a large heightmap in order. It is worth noting that we do not use postprocessing technology when comparing the performance with other state-of-the-art methods. In Section IV-E4, we compare the performance of EH before and after adding postprocessing technology.

B. Task-Specific Distillation

When the traditional MTL methods based on HPS deal with the multitasks with weak correlation, the shared information extracted from the backbone will affect the performance of the model, which leads to the phenomenon that one task becomes more generalized and the other weaker. Therefore, the methods based on SPS arise spontaneously, where each task has its own parameters and encourages parameter similarity by constraining the differences between the parameters of different tasks. Since each task is assigned an independent network, this method fails to capture commonalities among tasks, but it brings twice the parameters and calculations. Our motivation is to capture not only shared features, but also information extracted by specific experts for different tasks, so each task needs an independent gating extractor.

In order to demonstrate this, a straightforward opinion is to allow a set of TSGUs to obtain task-specific features from different stages. The upper and lower branches of TSP are responsible for distilling corresponding features from the shared features in the backbone network, as illustrated in Fig. 3. Given N tasks, the encoder of the proposed network is composed of a task-shared backbone represented as a function of $\mathcal{B}$ and N TSP networks $\mathcal{G}_k$ ($k = 2, 3, \ldots, N$). The task-shared backbone follows the input source $\mathbf{S}$, while the task-specific networks are based on the output of the shared network at the corresponding layer. For task k, the individual output $\mathcal{T}_k$ can be computed as

$$\mathcal{T}_k = \mathcal{G}_k(\mathcal{B}(\mathbf{S})). \tag{3}$$

As illustrated in Fig. 3, the first TSGU only takes the features in the task-shared backbone as input, while the input of the rest includes not only task-shared features, but also task-specific features of the upper stage. In other words, we extract the task-specific features from the backbone network at different stages and then relearn and align the features. Particularly, the gated distillation process is controlled by a set of mask maps, which themselves are estimated with one gating mask per channel to become a feature selector for the task-shared backbone. Therefore, the aporia of this section is how to achieve appropriate gating selectors for different tasks.

We represent the shared feature in stage i of the task-shared backbone as a^i, the task-specific feature of the task k in stage i as ${b_k}^i$, and the specific feature of the task k at the upper stage as ${b_k}^{i-1}$. The learned gating mask ${m_k}^i$ can be expressed as

$${m_k}^i = \begin{cases} \mathcal{S}\left({\mathcal{H}_k}^i\left(a^i\right)\right), & i = 1 \\ \mathcal{S}\left({\mathcal{H}_k}^i\left(\left[a^i, \left({b_k}^{i-1}\right)\right]\right)\right), & i \geq 2. \end{cases} \tag{4}$$

Here, $[\cdot, \cdot]$ denotes concatenation, $\mathcal{S}(\cdot)$ denotes a sigmoid activation function, and ${\mathcal{H}_k}^i(\cdot)$ indicates a bottleneck convolution block, as shown in Fig. 4. Correspondingly, the specific features in the current stage ${b_k}^i$ for task k can be obtained by first multiplying the shared feature a^i by the gating mask ${m_k}^i$ in an element-wise multiplication manner and then adopting an aligned convolutional block $\mathcal{F}^i(\cdot)$, consisting of a 1×1 convolution layer, batch normalization layer, and a sampling layer

$${b_k}^i = \mathcal{F}^i\left(a^i \otimes {m_k}^i\right), \quad i \geq 1 \tag{5}$$

where ${m_k}^i \in [0, 1]$ is automatically learned. Like a feature selector, the gating mask ${m_k}^i$ assists the distillation module to learn task-specific features. In fact, we can see that there are a significant difference in the gating mask between the two tasks during the course of the experiment. Detailed analysis will be shown in Section IV.

C. Cross-Task Propagation

Although the encoder-focused TSP architecture has achieved established achievements in MTL, due to limited memory resources, only limited local information can be exploited to generate gating masks. Therefore, the commonalities and differences between different tasks have not been fully explored, which may be beneficial to each other (e.g., the changes in semantic categories usually correspond to transitions in height values and vice versa). The specific information for task-level interaction was not clarified in detail in the previous cross-task fusion framework. To alleviate this issue, we utilize the generated features from the TSP network to improve each task output by introducing a decoder-focused CTP architecture.

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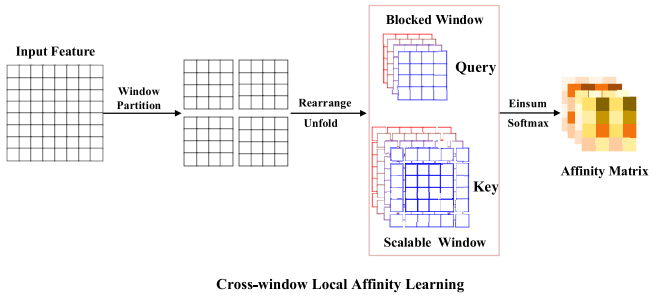


Fig. 5. Diagram of the cross-window local affinity learning layer (CwLAL-Layer).

1) Cross-Window Local Affinity Learning: The affinity matrix describing pair-wise similarities between pixels is formulated to model the common pattern relationships for the specific task. Due to the high-resolution input, the nonlocal affinity [12] computes the global relationship between a token and all other tokens in the dense prediction tasks, which suffers severely from serious quadratic complexity. Given a multichannel feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, the complexity of a global affinity matrix on $\mathbf{X}$ is $\mathcal{O}(H^2W^2C)$. Recently, a lot of work has been devoted to model the affinitive patterns among tasks in the local window and then diffuse them from the local region to the global region for efficient modeling. Although the window-based local affinity is computation friendly, the feature is divided in a nonoverlapping manner, which results in less information communication across subwindows. A straightforward approach in PSDNet [13] collects a $l \times l$ subwindow around each position, which can quickly lead to an out-of-memory exception. This approach could not be utilized to speed up practice.

Different from PSDNet [13], our method is motivated by the fact that neighboring positions share most of their neighborhood. Therefore, we pay attention to extracting pattern dependence from a local neighborhood for a block of pixels, rather than extracting individual neighborhoods for each position. To efficiently build the cross-window connections and diffuse the local affinity to global regions better, we focus on constructing high-order graphlets at less cost. Inspired by the intuitive form of local self-attention developed in HaloNet [21], we introduce a cross-window local affinity learning (CwLAL) mechanism to construct high-order graphlets, as shown in Fig. 5. We learn the local affinities across windows to avoid the limited receptive field caused by nonoverlapping windows. In this section, we utilize the pixel-level correlation to describe each local pattern, which is known as local affinity.

Specifically, given a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, we first divide it into N subwindows with the size of $l \times l$ in a nonoverlapping manner. Then we rearrange the subwindows as a group of query pixels to obtain the corresponding blocked windows. In addition, we extend the surrounding pixels to form the corresponding shared scalable window of shape $(l + 2h, l + 2h)$. Here, we call h scalable size. Finally, each scalable window behaving as key and blocked window behaving as query are executed Einsum and Softmax operation in parallel to obtain the corresponding local affinity matrix. For

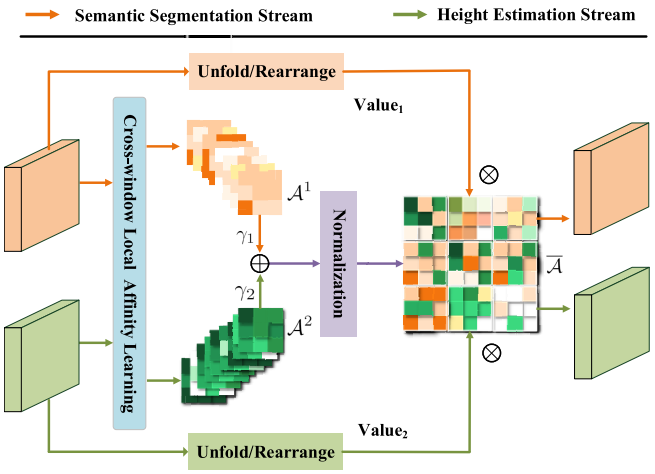


Fig. 6. Overview of the proposed MAD block. Local modality affinity can be adaptively transferred across tasks by adopting modality-affinitive matrix $\overline{\mathcal{A}}$. Here, $\bigotimes$ represents the matrix multiplication and $\bigoplus$ represent the weighted sum with two parameters γ_1 and γ_2.

the ith square region $\mathbf{F}_i \in \mathbb{R}^{C\times l\times l}$, we construct a graphlet $\mathcal{G}_i = \{\mathcal{B}_i, \mathcal{S}_i, \mathcal{A}_i\}$, where $\mathcal{B}_i$ and $\mathcal{S}_i$ indicate the feature matrix from the blocked window and the corresponding feature matrix from the scalable window separately. $\mathcal{A}_i$ behaving as affinity matrix can be defined formally as

$$[\mathcal{A}_i] = \sum [\mathcal{A}_i]_{\mathrm{ab}} = \sum \exp\left\{-\frac{\|[\mathcal{B}_\mathrm{i}]_\mathrm{a} - [\mathcal{S}_\mathrm{i}]_\mathrm{b}\|^2}{\delta^2}\right\} \tag{6}$$

in which $[\mathcal{A}_i]_{\mathrm{ab}}$ calculates the element correlation between the position a and b between the local region $\mathbf{B}_i$ and the shared neighborhood $\mathbf{S}_i$, and δ^2 (2 by default) is a scaling factor. For the whole graph, the global affinity matrix composed of N subwindows $\mathcal{A}_i$ can be written as

$$\mathcal{A} = \begin{bmatrix} [\mathcal{A}_1]^T & \cdots & 0 \\ 0 & [\mathcal{A}_2]^T & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & [\mathcal{A}_\mathrm{N}]^T \end{bmatrix} \tag{7}$$

where the function $(\cdot)^T$ indicates transpose operation and $\mathcal{A}$ is a matrix with zero elements except the main diagonal (nondiagonal matrix). Obviously, each local affinity records the relationships among $l \times l$ elements and $(l+2h) \times (l+2h)$ elements with computation cost $\mathcal{O}(l^2(l+2h)^2C)$, and the total cost of the nonlocal affinity matrix is $\mathcal{O}(Nl^2(l + 2h)^2C) = \mathcal{O}(HW(l + 2h)^2C) \ll \mathcal{O}(H^2W^2C)$, where $(l + 2h) \ll H$ and $(l + 2h) \ll W$. We utilize CwLAL-Layer to efficiently construct high-order graphlets across two tasks to mine the local patterns at the common locations.

2) Modality-Affinitive Diffusion: For MTL, there are commonalities that are beneficial to each other (e.g., the changes in semantic categories usually correspond to transitions in height values and vice versa). After the local affinity learning layer, we get a set of affinity matrices, which describe the pattern dependency among a local neighborhood of pixels. As we described earlier, different tasks have similar local affinity subgraphs in the same location, which implies that there

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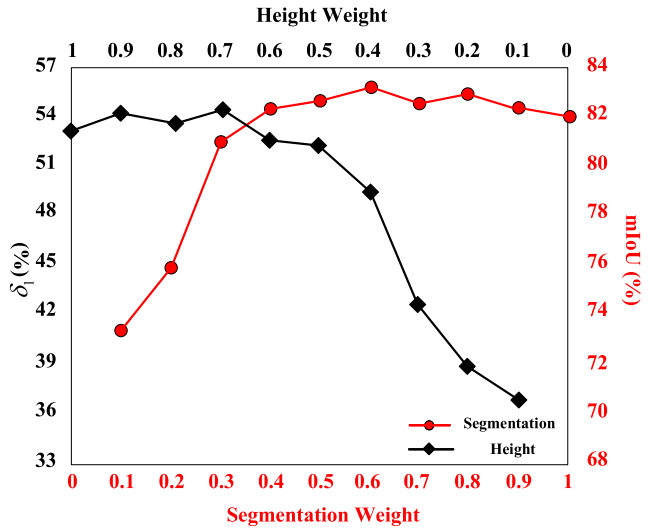


Fig. 7. Influence of weight selection on multitask performance. This figure illustrates the advantages of MTL in SS and EH, but manually adjusting the loss weight is tedious.

are some potential clues to explore the relationship between different tasks.

As illustrated in Fig. 6, we spread the main procedure of MAD between different tasks. First, the affinity matrix is learned through the local affinity learning layer to express the pairwise similarity of local pixels of each task. Note that, we did not add any supervision to learn affinity matrix $\mathcal{A}$ in order to save extra memory. Next, we want to integrate the interaction information for different tasks. We combine the local affinity matrixes $\mathcal{A}^1$ from task $\mathcal{T}_1$ and $\mathcal{A}^2$ of task $\mathcal{T}_2$ adaptively and assign different tasks a learnable weight

$$\overline{\mathcal{A}} = \gamma_1 \cdot \mathcal{A}^1 + \gamma_2 \cdot \mathcal{A}^2. \tag{8}$$

Finally, similar to the key in the previous section, each scalable window from different tasks behaves as value to perform a complete cross-task transfer process. By doing so, the commonalities and differences among tasks can be captured to further improve the output of each task in a one-time or recursive manner. Since we propose cross-window local affinity propagation, we adopt multiple iterations to acquire the long-range context information within each task. The detailed experimental analysis is shown in Section IV.

D. Model Objective With Dynamic Weighted Geometric Mean Strategy

In the previous part, we introduced the encoder–decoder framework for MTL. Nevertheless, one of the major challenges in MTL is the optimization process itself. Due to the case that one or more tasks play a leading role in the process of MTL, we have to consider the balance of multitask joint learning attentively. In this part, we introduce a novel dynamic weighted geometric averaging strategy for solving the task balancing problem.

1) Task-Specific Objective Function: There are two joint tasks, corresponding to SS and EH. In multiple pixel-wise tasks, each representation from input source $\mathbf{S}$ to task-specific targets $\mathbf{T}_i$ $(i = 2, 3, \ldots, N)$ is considered as one task. We optimize the classical objectives function designed for different tasks, where $\mathbf{T}$ represents the ground-truth target and $\overline{\mathbf{T}}$ indicates the output of the prediction.

1) For SS, we utilize a weighted cross-entropy loss function as optimization function to solve the problem of semantic category imbalance. More specifically, the inverse category frequency is considered as a balance factor for each pixel

$$\mathcal{L}_s(\mathbf{S}, \mathbf{T}_1) = -\frac{1}{mn}\sum_{m,n} w(m,n)\mathbf{T}_1(m,n)\log\overline{\mathbf{T}}_1(m,n). \tag{9}$$

2) For EH, we utilize an L2-Normalization to compare the predicted and ground-truth height. We apply the mean square error (MSE) of the predicted height and ground-truth as optimization function

$$\mathcal{L}_h(\mathbf{S}, \mathbf{T}_2) = -\frac{1}{mn}\sum_{m,n}\left[\mathbf{T}_2(m,n) - \overline{\mathbf{T}}_2(m,n)\right]^2. \tag{10}$$

The objective function provided above is only for specific tasks. However, how to find the right balance between these tasks is still a big challenge in multitask network training.

2) Multitask Optimization Function: Without loss of generality, the optimization objective function of MTL with N tasks, input source $\mathbf{S}$, and a set of task-specific targets $\mathbf{T}_i$ $(i = 2, 3, \ldots, N)$ can be expressed as

$$\mathcal{L}_{\text{MTL}}(\mathbf{S}, \mathbf{T}) = \sum_{i}^{N} w_i \mathcal{L}_i(\mathbf{S}, \mathbf{T}_i). \tag{11}$$

Equation (11) indicates the linear combination of task-specific losses $\mathcal{L}_i$ and w_i controls the relative weights for specific tasks, respectively. Compared with the direct addition of losses $\mathcal{L}_i$, this method allows us to manually adjust the weight of each task. As illustrated in Fig. 7, the multitask performance is extremely sensitive to w_i. However, the weight selection of w_i is hard to tune manually and each training usually takes a lot of time. Since different tasks may be in different learning stages, the fixed w_i that always accompanies the whole training process will limit the task learning at some stage. In addition, this method suffers from different scales (order of magnitude) of individual loss function during backpropagated gradients. Therefore, the better weighting method should be dynamic, which tunes weight hyper-parameters according to the learning stages of different tasks.

According to the definition and characteristics of geometric mean [41] and arithmetic mean [42], we can find that arithmetic mean is more affected by extreme value than geometric mean, which may be more robust when extreme value exists in the samples. In addition, we find that the geometric mean is not very sensitive to the magnitude of the value, which is very important for multitask optimization. In this article, a DWGeoMean method is explored to replace the traditional arithmetic mean combination method. We refer to this as dynamic weighted geometric mean strategy (DWGMS), which

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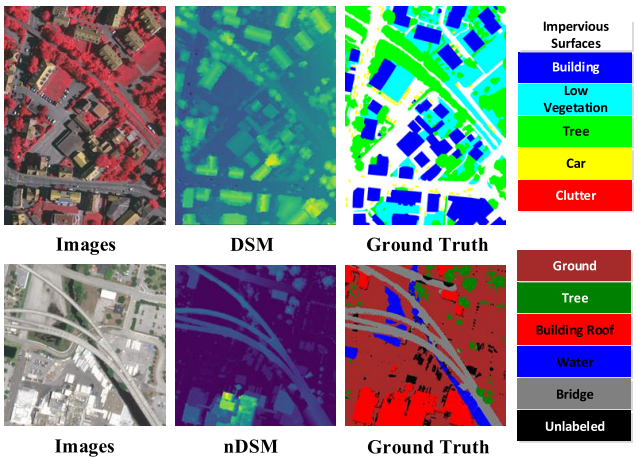


Fig. 8. Experimental datasets. The upper, respectively, represents the original images, DSM, ground-truth labels, and the corresponding color codes for the Vaihingen dataset. The lower, respectively, indicates the original images, normalized digital surface model (nDSM), ground-truth labels, and the corresponding color codes for the US3D dataset.

can be written as

$$\mathcal{L}_{\mathrm{MTL}}(\mathbf{S},\mathbf{T},t)=\prod_{i=1}^{N}\sqrt[\sum w_i(t)]{\mathcal{L}_i(\mathbf{S},\mathbf{T}_i)^{w_i(t)}}. \tag{12}$$

Equation (12) expounds the DWGeoMean procedure of task-specific losses $\mathcal{L}_i$, in which $w_i(t)$ is a dynamic and self-learning hyperparameter. Most importantly, this function is differentiable and ensures that all tasks can achieve high performance. Here, considering the two-task problem for SS $\mathbf{T}_1$ and EH $\mathbf{T}_2$, we define our optimization objective function as

$$\mathcal{L}_{\mathrm{MTL}}(\mathbf{S},\mathbf{T},t)=\mathcal{L}_s(\mathbf{S},\mathbf{T}_1)^{w(t)}\mathcal{L}_h(\mathbf{S},\mathbf{T}_2)^{1-w(t)}. \tag{13}$$

Here, we pay unequal attention to different tasks by introducing $w(t)$, whose initialization is set to 0.5 by default. During the process of multitask training, we do not need to fit the variance [18] and calculate global gradient [19] as well as metrics for different steps [11]. We can simply implement this new loss function for multitask training without increasing the complexity.

IV. Experiments

A. Datasets

In order to evaluate the effectiveness of the proposed framework for MTL, we conduct extensive experiments on aerial and satellite image datasets, as illustrated in Fig. 8. The upper represents ISPRS Vaihingen [43] aerial images, which contain very high-resolution images from Germany. And the lower indicates the US3D dataset [44] evaluated as the satellite images, which covers approximately 100 square kilometers of the United States. For each image in the above datasets, the corresponding elevation data and ground-truth labels are provided.

1) Aerial and Satellite Images:

1) *ISPRS Vaihingen:* The Vaihingen dataset [43] contains 33 patches, where the training set embodies 16 tiles and the remaining 17 tiles are selected to test the proposed method. Each aerial image contains the corresponding true orthophoto (TOP), semantic label, and DSM, which has an average size of 2494 × 2064 pixels and a ground sampling distance of 9 cm.
2) *US3D:* The US3D dataset [44] contains 2833 patches consisting of WorldView-3 RGB images and normalized digital surface models (nDSMs) with a spatial resolution of 30 cm/pixel and a size of 1024 × 1024 pixels. Following the previous works, 2783 tiles with corresponding ground-truth labels are selected as training sets, and the remaining 50 tiles are selected to test the proposed network. It is noteworthy that the multispectral images are not utilized in our experiments.

B. Evaluation Metrics

To comprehensively estimate the performance of our proposed MTL method, we quantify the performance of the model by a total of nine common evaluation metrics from two aspects: SS and EH.

1) Semantic Segmentation: For SS, we employ intersection over union (IoU), overall accuracy (OA), and $F1$ score to validate the performance of our model

$$\mathrm{IoU}=\frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}}+N_{\mathrm{FP}}+N_{\mathrm{FN}}} \tag{14}$$

$$\mathrm{OA}=\frac{N_{\mathrm{TP}}+N_{\mathrm{TN}}}{N_{\mathrm{TP}}+N_{\mathrm{FP}}+N_{\mathrm{FN}}+N_{\mathrm{TN}}} \tag{15}$$

$$F_1=\frac{2\cdot\mathrm{precision}\cdot\mathrm{recall}}{\mathrm{precision}+\mathrm{recall}} \tag{16}$$

where N_{TP}, N_{FP}, N_{TN}, and N_{FN} are the number of true positives, false positives, true negatives, and false negatives, respectively. And precision $=(N_{\mathrm{TP}}/N_{\mathrm{TP}}+N_{\mathrm{FP}})$ and recall $=(N_{\mathrm{TP}}/N_{\mathrm{TP}}+N_{\mathrm{FN}})$.

2) Height Estimation: For EH, we adopt root mean square error (RMSE), average relative error (Rel), as well as lg(RMSE) to clarify the accuracy of the proposed approach, following [8]:

$$\mathrm{Rel}=\frac{1}{n}\sum\frac{|H_g-H_e|}{H_g} \tag{17}$$

$$\mathrm{RMSE}=\sqrt{\frac{1}{n}\sum\left(H_g-H_e\right)^2} \tag{18}$$

$$\mathrm{lg(RMSE)}=\sqrt{\frac{1}{n}\sum\left(|\lg H_g-\lg H_e|\right)^2} \tag{19}$$

where H_g and H_e indicate the ground-truth height and the estimated height, separately. n is the total number of pixels in the image. Additionally, we adopt the ratio of pixels to show how close the estimated height is to the ground-truth height, which can be defined as

$$\delta_i=\max\left(\frac{H_g}{H_e},\frac{H_e}{H_g}\right)<1.25^i,\quad i\in\{1,2,3\}. \tag{20}$$

From (17) to (19), the criterion defined by formula represents the error value, while the criterion defined by (20) represents the accuracy of prediction height. Therefore, the lower the value, the better for the former, the higher the value, the better for the latter.

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TABLE I

QUANTITATIVE PERFORMANCE FOR SS AND EH ON THE VAIHINGEN TEST SET. "SS" AND "HE," RESPECTIVELY, INDICATE SS AND EH AS STL METHODS. "MTL" INDICATES MULTITASK LEARNING ARCHITECTURES. THE BOLD ENTITIES REPRESENT THE BEST PERFORMANCE

	Model	T1:Semantic Seg.			T2:Height Est.					
		(Higher Better)(%)			(Lower Better)			(Higher Better)		
		mean F_1	**OA**	**mIoU**	**Rel**	**Rmse**	**lg(Rmse)**	δ_1	δ_2	δ_3
SS	FCN [23]	83.74	86.51	72.69	-	-	-	-	-	-
	Deeplab V3 [48]	89.57	90.56	81.47	-	-	-	-	-	-
	PSPNet [24]	90.26	90.85	82.58	-	-	-	-	-	-
	CASIA2 [49]	90.10	91.10	-	-	-	-	-	-	-
	GANet [50]	90.20	91.10	-	-	-	-	-	-	-
HE	Amirkolaee *et al.*[8]	-	-	-	1.163	2.871	0.334	0.330	0.572	0.741
	IMG2DSM [37]	-	-	-	-	2.58 ± 0.09	-	-	-	-
	D3Net [51]	-	-	-	2.016	2.123	-	0.369	0.533	0.644
	Li *et al.*[9]	-	-	-	0.314	1.698	0.155	0.451	0.817	0.939
MTL	BAML[52]	86.90	88.40	-	1.064	1.762	-	0.451	0.617	0.714
	Cross-Stitch [16]	89.43	90.48	81.38	0.434	2.189	0.244	0.394	0.563	0.747
	PSDNet [13]	90.21	90.72	82.38	0.363	1.542	0.171	0.447	0.745	0.906
	MTAN [11]	90.26	90.71	82.48	0.258	1.276	0.143	0.533	0.810	0.952
	ASSEH(ResNet-50)	89.84	90.37	81.80	0.272	1.380	0.146	0.468	0.788	0.951
	ASSEH(ResNet-101)	**90.85**	**91.42**	**83.41**	**0.237**	**1.144**	**0.120**	**0.595**	**0.860**	**0.971**

C. Implementation Details

The proposed network is implemented with the PyTorch platform in a batch size of 4 on a single Tesla P100 graphics processing unit (GPU). We adopt ResNet-50 or ResNet-101 [38] trained on the ImageNet dataset [45] as the shared backbone to extract features and the multitask decoders are initialized randomly. Following the previous works [25], [29], our MTL network is optimized by an SGD [46] with weight decay 0.0005 and momentum 0.9. The initial learning rate is set to 0.01 and is multiplied by $1 - ((\text{iter}/\text{max_iter}))^{\text{power}}$ with power $=$ 0.9. In addition, InPlace-ABNSync [47] is employed to synchronize the mean and standard deviation of multiple branches. For data augmentation, we crop the images randomly into 512×512 during the training process. In the test phase, we improve the test performance of aerial and satellite images by adopting multiscale inputs.

D. Experimental Results

1) Results on Vaihingen: The experimental results on the Vaihingen dataset are reported in Table I, where the proposed ASSEH model is compared with other state-of-the-arts. Note that, for SS, we calculate the mean F_1 and mIoU values for all classes except clutter/background and count the OA value for all categories, following most methods [1], [49], [50]. In this table, we compare the performance of the method from the perspective of STL and MTL. For SS, our ASSEH (ResNet-101) achieves ascendant performance than other methods with mean $\boldsymbol{F_1}$ of 90.85%, OA of 91.42%, and mIoU of 83.41% by adopting multiscale inference (MSI). For EH, the proposed ASSEH model achieves 0.237 in Rel, 1.144 m in Rmse, 0.120 in lg(Rmse), and a better accuracy in δ_1, δ_2, δ_3. It also shows that the classification and regression prediction performance of our model are not affected by MTL. On the contrary, they benefit from the commonalities and differences between multitasks.

Through visualizations in Fig. 9, we can comprehend the advantages of our MTL method over the ordinary STL. For pixel-level segmentation tasks, the boundaries of the objects are more perfect and evident, benefiting from the introduction of 3-D height information. In addition, the proposed model can better distinguish the classes with similar spectral properties, as well as tree and low vegetation, building, and impervious surfaces. For pixel-level EH, the regression height values within the region tend to be smooth, profiting from context information from SS. The brighter the color, the higher the height value of the position.

2) Results on US3D: Different from the Vaihingen dataset, the satellite dataset US3D with lower spatial resolution brings greater challenges to MTL. We report the quantitative performance of our ASSEH and the comparing state-of-the-arts on the US3D dataset in Table II from the perspective of STL and MTL. Note that we evaluate the proposed method without MSI since it brings no performance improvement for this challenge. As illustrated in Table II, the proposed ASSEH method achieves ascendant performance than all comparing single-task or multitask models on mean $\boldsymbol{F_1}$, OA, and mIoU for SS. On the other hand, our ASSEH model achieves the best on the metrics, such as Rel, lg(Rmse), δ_1, δ_2, and δ_3 for EH. The experimental results also indicate that the related tasks can enhance the generalization performance of the model on each task by using the domain knowledge contained in the supervision signal of related tasks.

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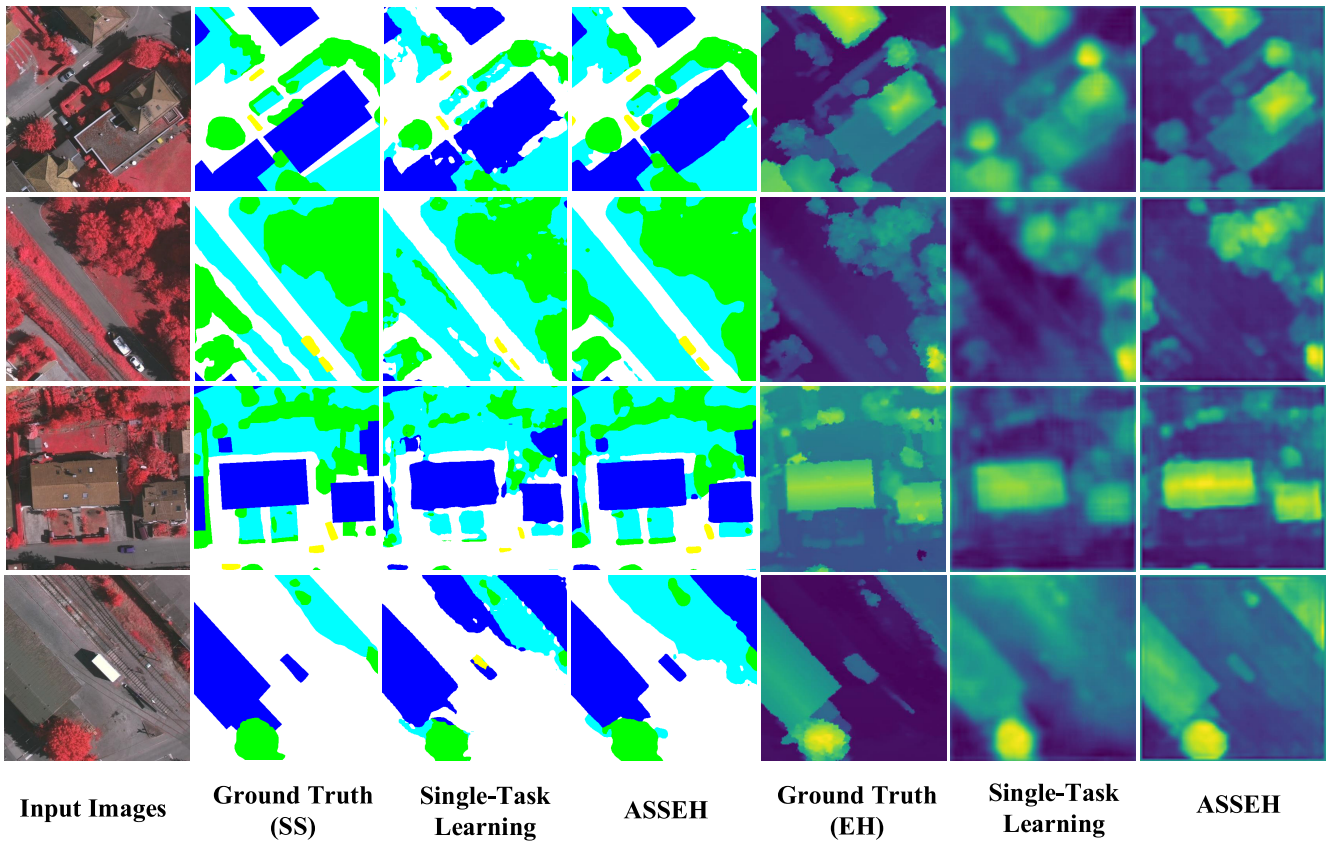


Fig. 9. Vaihingen qualitative results on six-class SS and EH. White: impervious surfaces, blue: buildings, cyan: low vegetation, green: trees, and yellow: cars. The input images are cropped to 512 × 512 for better visualization.

TABLE II

QUANTITATIVE PERFORMANCE FOR SS AND EH ON THE US3D TEST SET. "SS" AND "HE," RESPECTIVELY, INDICATE SS AND EH AS STL METHODS. "MTL" INDICATES MULTITASK LEARNING ARCHITECTURES. THE BOLD ENTITIES REPRESENT THE BEST PERFORMANCE

	Model	T1:Semantic Seg.			T2:Height Est.				
		(Higher Better)			(Lower Better)		(Higher Better)		
		mean F_1(%)	**OA(%)**	**mIoU(%)**	**Rel**	**lg(Rmse)**	δ_1	δ_2	δ_3
SS	FCN [23]	71.04	81.92	58.60	-	-	-	-	-
	Deeplab V3 [48]	79.91	85.24	69.21	-	-	-	-	-
	PSPNet [24]	81.86	85.74	70.53	-	-	-	-	-
	GANet [50]	83.04	86.42	71.21	-	-	-	-	-
HE	D3Net [51]	-	-	-	0.526	0.208	0.256	0.635	0.846
	DORN [53]	-	-	-	0.488	0.200	0.317	0.646	0.859
	FastDepth [54]	-	-	-	0.383	0.189	0.384	0.701	0.875
MTL	Cross-Stitch [16]	77.53	84.06	67.52	0.535	0.222	0.217	0.625	0.834
	PSDNet [13]	82.34	86.27	70.81	0.429	0.192	0.375	0.706	0.891
	MTAN [11]	82.55	86.36	70.82	0.390	0.186	0.406	0.709	0.887
	ASSEH(ResNet-50)	80.47	85.16	69.36	0.491	0.206	0.334	0.654	0.863
	ASSEH(ResNet-101)	**83.81**	**86.99**	**72.65**	**0.339**	**0.173**	**0.448**	**0.753**	**0.905**

We conduct qualitative and visual analyses of the experimental results of the US3D dataset, as illustrated in Fig. 10. In the inference phase, our ASSEH method only takes optical images as input but can obtain simultaneous segmentation and height maps at the same time. It can be clearly observed that our ASSEH method produces desirable segmentation results and more refined height maps. We attribute this to the joint MTL and the interaction of 2-D contextual and 3-D geometric information. Specifically, for height-aware SS, ASSEH performs better in complex categories as the objects

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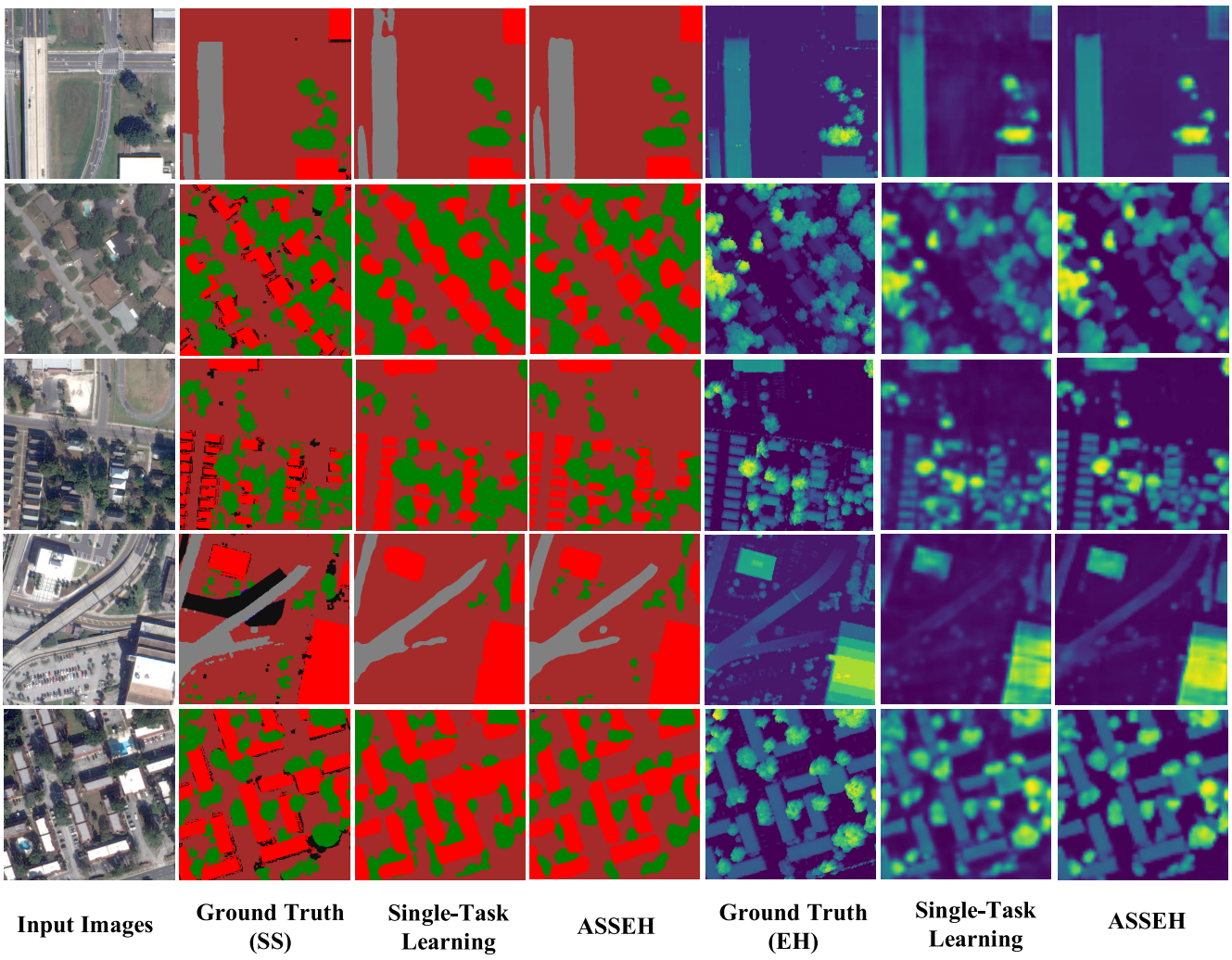


Fig. 10. US3D qualitative results on six-class SS and EH. The input images are cropped to 512 × 512 for better visualization.

with similar spectral information can be well distinguished by introducing geometric information. For semantic-aware EH, per class height changes are consistent with our observations as semantic awareness helps to distinguish different objects in high space (e.g., ground, tree, and building). The brighter the color, the higher the height value of the position.

E. Ablation Analysis

In this section, extensive ablation experiments have been carried out on Vaihingen validation sets to verify the effects of key components of our ASSEH model and model objective function.

1) Analysis on Network Settings: We first analyze the benefit of associatively estimating height and segmenting semantics by adopting our proposed TSD module, CTP module, and DWGeoMean strategy. Here, the vanilla FCN [23] based on ResNet-101 [38] is applied for single-task learning. Taking the STL model as a baseline, we separately train two models for SS and EH tasks. Compared with the proposed ASSEH, the STL model lacks the TSP module and the CTP module. For equally comparing, we similarly select ResNet-101 [38] as the shared backbone in ASSEH and ensure all training and testing procedures in each ablation experiment are consistent. As illustrated in Table III, the multitask model has a better performance than the single-task model, and further multitask model using our TSD, CTP, and DWGeoMean obtains the best results. The advantage of the proposed method over the STL model can give the credit to the generalization ability of MTL, while the over-traditional MTL model proves the effectiveness of the proposed modules.

Specifically, we can comprehend that MTL with the DWGeoMean strategy outperforms other MTL with fixed and uncertainty strategies. This is mainly attributed to the fact that the loss combination function of the two methods is the arithmetic mean, and they suffer from differences in the magnitudes of the individual losses. However, the DWGeoMean strategy is not only insensitive to the magnitude of the individual losses, but also can adaptively adjust the weight of the loss function. In addition, we plot the OA for each category in the SS task and multiple evaluation performances for the EH task as a radar chart in Fig. 11. In short, it can be clearly aware that each module or method proposed by us does promote each task in the process of associative learning.

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TABLE III

ABLATION STUDY OF THE PROPOSED MODEL ON VAIHINGEN TEST SET. THE BEST PERFORMANCE COMBINATIONS ARE HIGHLIGHTED IN BOLD. THE HIGHEST VERIFICATION SCORE FOR EACH TASK IS INDICATED BY A BOX

Model	Weighting	Semantic Seg.		Height Est.	
		OA	mIoU	Rmse	δ_1
STL	N.A.	89.98	79.53	2.236	0.349
MTL	Fixed	89.43	78.86	2.471	0.325
	Uncertainty	**90.38**	**81.44**	2.140	0.353
	DWGeoMean	90.23	81.16	**1.932**	**0.377**
MTL+TSD	Fixed	90.53	81.83	1.746	0.416
	Uncertainty	90.67	82.10	**1.395**	**0.517**
	DWGeoMean	**90.74**	**82.39**	1.568	0.459
MTL+TSD+CTP	Fixed	90.85	82.71	1.364	0.521
	Uncertainty	91.01	82.94	1.276	0.563
	DWGeoMean	**91.42**	**83.41**	**1.144**	**0.595**

TABLE IV

COMPARISONS AMONG MSI AND MULTIGRID STRUCTURE (MGS) STRATEGIES ON VAIHINGEN TEST SET

Model		Semantic Seg.		Height Est.	
MSI	MGS	OA	mIoU	Rmse	δ_1
✗	✗	90.93	82.83	1.173	0.584
✗	✓	91.04	83.08	1.156	0.591
✓	✗	91.26	83.21	1.162	0.589
✓	✓	**91.42**	**83.41**	**1.144**	**0.595**

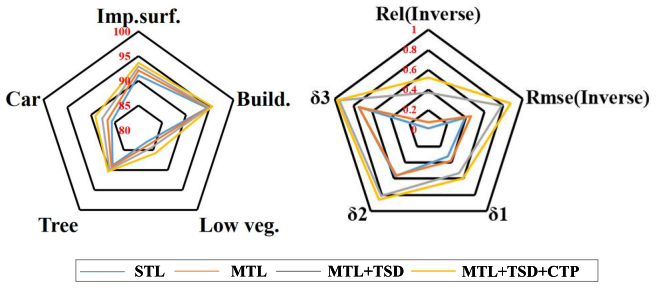


Fig. 11. Radar charts of qualitative results on SS and EH, trained with the DWGeoMean strategy. High is better. Left is the percentage OA per category in the SS task, while the right is the evaluation result per attribute for the EH task. Here, both Rel and Rmse are inversed, and Rel(Inverse) is normalized for comparison. Zoomed-in view for details.

In addition, we adopt a series of measures for better performance following [27], such as MSI and multigrid structure (MGS). Significantly, we average the segmentation results with five scales {0.5, 0.75, 1.0, 1.25, 1.5} for inference. Meanwhile, we apply a structure of grids with multiple sizes {1, 2, 4} in the last layer. As illustrated in Table IV, all the above strategies have significantly improved performance.

2) Effect of Task-Specific Distillation Module: In order to verify the effectiveness of the proposed TSP module, we carry out quantitative and qualitative verification analyses

TABLE V

PERFORMANCE COMPARISON WITH TRADITIONAL HPS AND SPS METHODS

Model	Semantic Seg.		Height Est.	
	OA	mIoU	Rmse	δ_1
MTL_HPS	89.43	78.86	2.471	0.325
MTL_SPS[16]	90.48	81.38	2.189	0.394
MTL_TSD	**90.74**	**82.39**	**1.568**	**0.459**

TABLE VI

EFFICIENCY COMPARISON WITH TRADITIONAL HPS AND SPS METHODS. HERE, EXECUTION TIME REFERS TO THE TIME OF INFERRING AN IMAGE OF SIZE [1 × 3 × 512 × 512]

Model	Params(M)	Execution time(s)
MTL_HPS	48.29	4.033
MTL_SPS[16]	105.43	12.767
MTL_TSD	77.37	6.538

on the Vaihingen dataset. Here, MTL_HPS refers to sharing a backbone network, followed by two task-specific decoders. MTL_SPS refers to learning a task-specific subnetwork for each task with some cross-stitch units [16] to access the information inside the other, followed by two decoders. MTL_TSD refers to the proposed network without a CTP module. For a fair comparison, all methods are compared under the same experimental settings.

In the case of quantitative analysis, we depict performances and efficiencies per method for a comprehensive comparison. As illustrated in Table V, the proposed TSD achieves the optimal performance during the two tasks compared with both HPS and soft-parameter sharing (SPS) methods. Another discovery shown in Table VI is that the parameters of our model are 1.6 times that of the HPS method, while the SPS method is more than twice that of the HPS method. In addition, we verify the running efficiency of the model by inferring a random tensor of size [1 × 3 × 512 × 512]. Although the introduction of the TSP module leads to the addition of model complexity and running efficiency, it greatly improves the performance of MTL. Such good results benefit from the task-shared features for generalization across tasks and the task-specific features to avoid under-fitting.

In the case of qualitative analysis, we visualize the middle-layer feature maps and the corresponding gating masks from the TSGU on the Vaihingen dataset to better understand the role of the proposed distillation module. As shown in Fig. 12, there is an obvious difference in the masks between SS and EH. Each mask serves as a feature selector to extract useful information for each task and suppresses useless parts of shared features. It is worth noting that the height mask has higher contrast than the semantic mask, which indicates that EH benefits more from task-specific features and SS benefits more from task-shared features.

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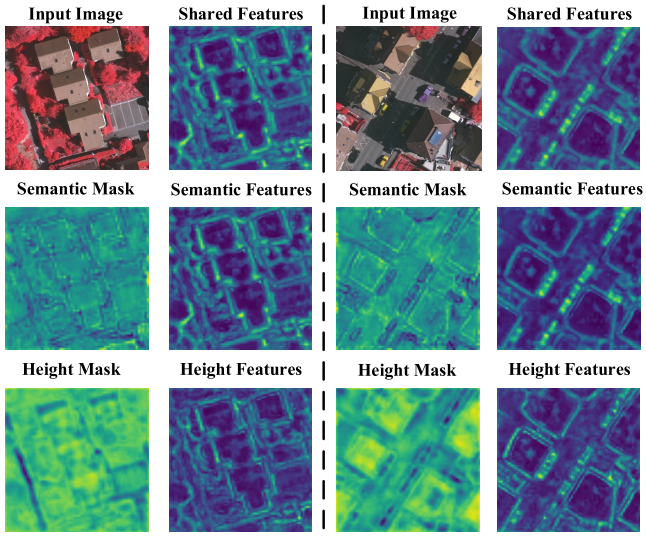


Fig. 12. Visualization of the semantic and height masks, as well as semantic and height features of the proposed network in the middle layer. Best view in color.

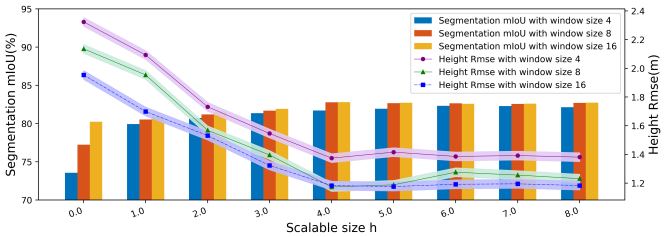


Fig. 13. Influence of the window size l and scalable size h. (Bars) Segmentation mIoU (left axis, higher the better) of ASSEH w.r.t. window size and (lines) height rmse (right axis, lower the better) of ASSEH w.r.t. window size. Best view in color.

3) Effect of Cross-Task Propagation Module: CTP module consists of a CwLAL layer and modality-affinity diffusion block, which aims at capturing the matched local affinity information between tasks. In this section, we conduct large quantities of experiments to verify the effectiveness of the CTP module from the following two aspects.

a) Analysis on window size l and scalable size h.: As mentioned in Section III-C, the receptive field of the CwLAL layer is $(l + 2h, l + 2h)$, where l is the window size and h is the scalable size. From Fig. 13, we can see two interesting phenomena as follows.

1) With the fixed scalable size, the performance of MTL mostly improves as the window size increases. Moreover, when the window size reaches 8, increasing the window size will bring less performance improvement but more extra computation. We analyze this mainly because with the increase of window size, broader local affinity is learned, which makes the relationship between pixels in the task more complex and is more sensitive to feature response.
2) With the fixed window size, the performance of MTL does not increase consistently with the increase of scalable size, but reaches the optimal value at the size of 4. We argue that the main reason for this problem is that some details may be blurred to a certain extent as the scalable size increases.

TABLE VII

RESULTS OF ITERATION NUMBERS OF THE CTP MODULE IN DIFFUSION PROCESS. THE PERFORMANCE AND TIME BURDEN CHANGES CAN BE SEEN AS A TRADEOFF

#CTP	Semantic Seg.		Height Est.		Times(ms)
	OA	mIoU	Rmse	δ_1	
0	90.74	82.39	1.568	0.459	**0**
3	90.81	82.66	1.431	0.516	93.23
6	91.03	83.17	1.247	0.564	102.66
9	**91.42**	**83.41**	**1.144**	**0.595**	109.03
12	91.25	83.24	1.206	0.579	116.44
15	90.96	82.92	1.351	0.513	125.05
18	90.86	82.75	1.463	0.496	132.02

TABLE VIII

ABLATION STUDY OF THE PROPOSED DWGEOMEAN STRATEGY ON THE VAIHINGEN TEST SET. HERE, WCE REPRESENTS THE WEIGHTED CROSS-ENTROPY LOSS FOR SS. L1 AND L2, RESPECTIVELY, REPRESENT MAE AND MSE LOSS FOR EH. THE BEST PERFORMANCE COMBINATIONS ARE HIGHLIGHTED IN BOLD. THE HIGHEST VERIFICATION SCORE FOR EACH TASK IS INDICATED BY A BOX

Model	Weighting	Semantic Seg.		Height Est.	
		OA	mIoU	Rmse	δ_1
wCE + L1	Fixed	90.74	82.65	1.594	0.426
	Uncertainty	91.11	83.12	1.582	0.439
	DWGeoMean	**91.32**	**83.28**	**1.354**	**0.515**
wCE + L2	Fixed	90.85	82.71	1.364	0.521
	Uncertainty	91.01	82.94	1.276	0.563
	DWGeoMean	**91.42**	**83.41**	**1.144**	**0.595**

b) Analysis on iteration numbers.: Here, we explore the optimal iteration number for the diffusion process. The whole MTL model is based on resnet-101 and learns cross-window local affinity on the 1/8 input scale with the window size and scalable size set to 8 and 4, respectively. As mentioned in Section III-C, we adopt multiple iterations to capture the long-distance pattern affinity within each task. From Table VII, we can observe that the performances of the two tasks are improved at first and gradually converge from the ninth iteration. These results reveal that regularization between tasks can be enhanced as the iteration number increases. However, the diffusion of a too long-distance affinity matrix may have a negative impact on the current regional model to a certain extent. We argue that the pattern affinity of each pixel in the pixel-level prediction task is highly similar to that of its neighbors compared to distant pixels. In addition, the testing time is carried out under the same experimental setting, such as the test image size (512×512) and other network settings. As illustrated in Table VII, the time increases with more iterations, which can be regarded as a tradeoff.

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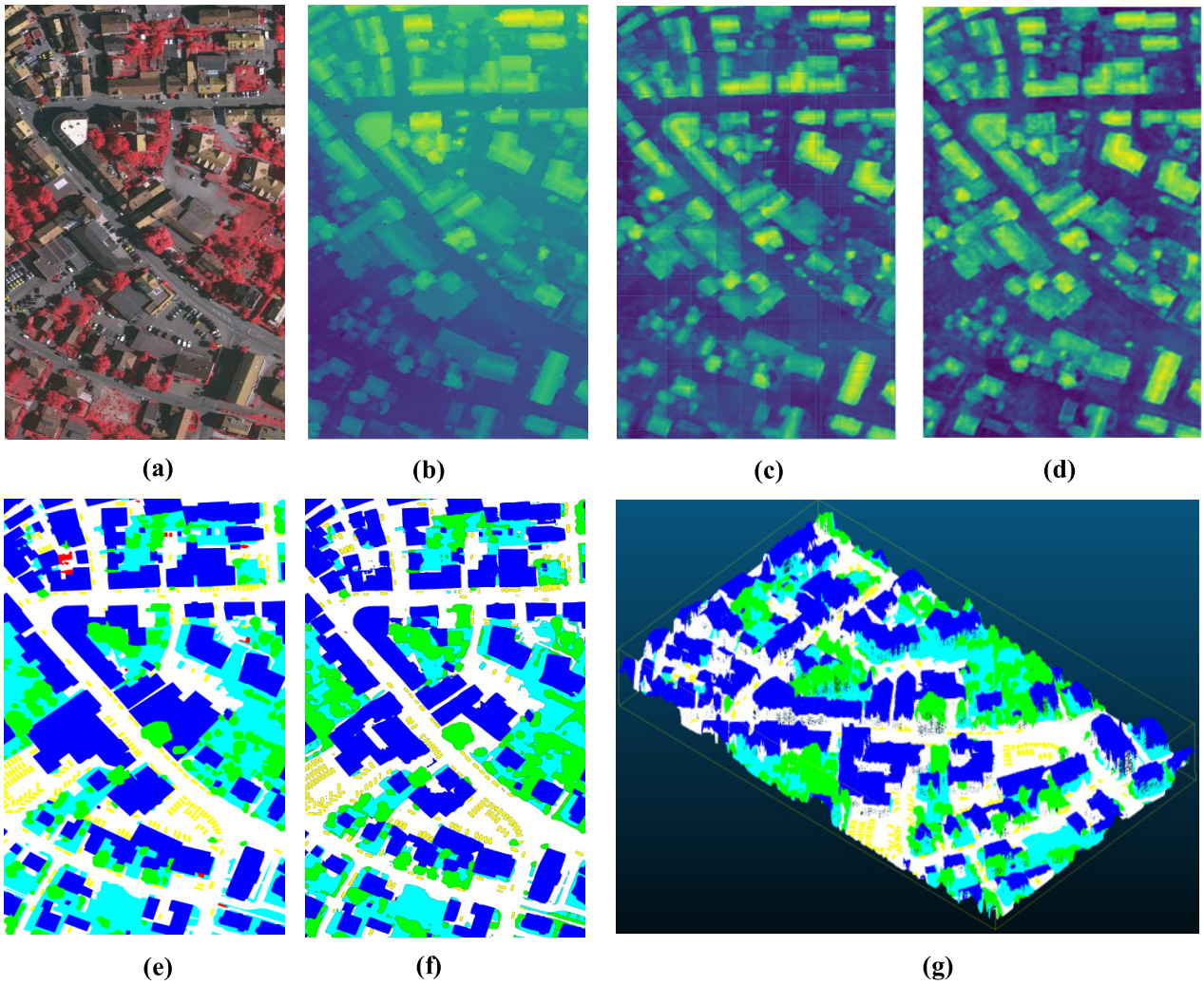


Fig. 14. Visualization results of EH with postprocessing technique (PPT) and semantic stereo reconstruction on the ISPRS Vaihingen dataset. (a) Images. (b) Ground truth (EH). (c) Patch-wise height map. (d) Seamless height map. (e) Ground truth (SS). (f) Segmentation map. (g) Semantic stereo reconstruction.

TABLE IX
PERFORMANCE COMPARISON OF EH WITH AND WITHOUT PPT ON THE VAIHINGEN TEST DATASET

Model	Height Est.			
	(Lower Better)		(Higher Better)	
	Rel	Rmse	δ_1	δ_3
w/o PPT	0.237	1.144	0.595	0.971
w/ PPT ($\mu = 0, \sigma = 0.1$)	0.241	1.150	0.591	0.968
w/ PPT ($\mu = 0, \sigma = 0.2$)	0.240	1.149	0.593	0.970
w/ PPT ($\mu = 0, \sigma = 0.5$)	0.239	1.145	0.595	0.972
w/ PPT ($\mu = 0, \sigma = 1$)	0.238	1.144	0.596	0.972
w/ PPT ($\mu = 0, \sigma = 2$)	0.237	1.144	0.595	0.971

4) Ablation Study of DWGeoMean Strategy: We conduct generous experiments by employing different loss functions, different combinations, and different weighting to verify the effectiveness of the DWGeoMean strategy. Here, "wCE + L1" means ASSEH employs the weighted cross-entropy loss for SS and mean absolute error (MAE) loss for EH. "wCE + L2" means ASSEH employs the weighted cross-entropy loss for SS and MSE loss for EH. As shown in Table VIII, DWGeoMean achieves better results than the other two when the loss functions with different magnitudes. The reasons are: the geometric mean is less sensitive to the magnitude of variables than the arithmetic mean. At the same time, the arithmetic mean is more affected by extreme values than the geometric mean.

5) Result of Postprocessing Technique (PPT) and Semantic Stereo Reconstruction: As shown in Fig. 14(c), connecting adjacent small height maps directly can cause gaps between adjacent patches. To alleviate the issue, we employ a postprocessing technique (PPT) to adjust the swing of the height value and shorten the height difference among adjacent patches. Note that we do not use postprocessing technology in all previous performance comparisons of high estimation. In order to verify the effectiveness of PPT, we compare the EH performance with and without PPT in Table IX. We find that with the increase of σ, most of the performance of EH first becomes worse and then gradually returns to the performance

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without PPT, which is due to the nature of the 2-D Gaussian smoothing kernel function. However, with the increase of σ, the gap between adjacent blocks still exists. We present a seamless height map by employing a 2-D Gaussian smoothing kernel function ($\mu = 0, \sigma = 0.2$), as illustrated in Fig. 14(d).

In addition, the SS and EH results are integrated pixel by pixel for semantic stereo reconstruction to enhance the interpretation ability of real remote-sensing scenes, as illustrated in Fig. 14(g).

V. Conclusion

In this article, a unified learning model of SS and EH is established to interpret the jointly 2-D contextual and 3-D geometric information region on multisource remote-sensing datasets. Particularly, our framework is encapsulated into encoder–decoder structure, consisting of a shared backbone, TSD module, CTP module, and two task-specific decoders. The TSD module embraces a set of TSGU for each task to allow the task-specific features to be tailored from the backbone and the task-shared features to be transmitted across tasks. It benefits from the task-shared features for generalization across tasks and the task-specific features to avoid under-fitting. Furthermore, the CTP module is proposed to mine common local patterns across tasks by constructing and diffusing the local pattern graphlets, which is beneficial to promoting task interaction. In addition, we design a DWGeoMean strategy, which can not only automatically learn the weights for each task, but also be more robust to the magnitude of the loss function. Compared with STL for each task, the proposed ASSEH can learn the features tailored for each task to enhance the generalization of the method and achieve better performance on each task. The experimental results on the challenging Vaihingen and US3D datasets demonstrate that our method is better than state-of-the-art SS and EH methods. In future work, we will further restrict the number of network parameters as much as possible as the number of tasks grows to be implemented in resource-limited applications such as unmanned aerial vehicles (UAVs) and satellite platforms.

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